

TEAM ASTERIX

Autonomous Conservation System (ACS)

"Ground when possible. Air when necessary. Disturb nature as little as possible."

Cooperative Ground-Air Swarm Platform for Wilderness Protection

aBAJA 2026 Engineering Competition Entry

The Reach vs. Ecological Impact Dilemma

- ✓ **Ground Patrolling Impact:** Heavy standard ATVs cause critical soil compaction, destroy sensitive forest root networks, and disturb wildlife feeding/mating beats with exhaust and loud engine noises.
- ✓ **Aerial Patrolling Limitations:** Commercial drones provide excellent line-of-sight reach but suffer from short flight times (20–30 mins), tiny payload capacities, and cannot take physical soil or river samples.
- ✓ **Isolation & Blindspots:** Thick forest foliage blocks GPS and radio lines of sight, leaving rangers blind to poaching or wild fires in deep valleys.



Zero-Disturbance

How do we protect deep forest regions without entering them with heavy machinery?

The Cooperative Ground-Air Swarm Solution

- ✓ **A Shared Ecosystem:** Moving away from a standalone buggy. We propose a mobile coordinate system combining an AEUV (ground hub) and EcoWing-U drones (aerial extensions).
- ✓ **Role of the AEUV:** Stays strictly on designated forest logging trails. Serves as a mobile charging grid, computing core, robotic sampler, and drone airport.
- ✓ **Role of the Drones:** Dispatched dynamically from the vehicle to deploy sensors, drop seeds, or administer vaccines in trackless valleys.

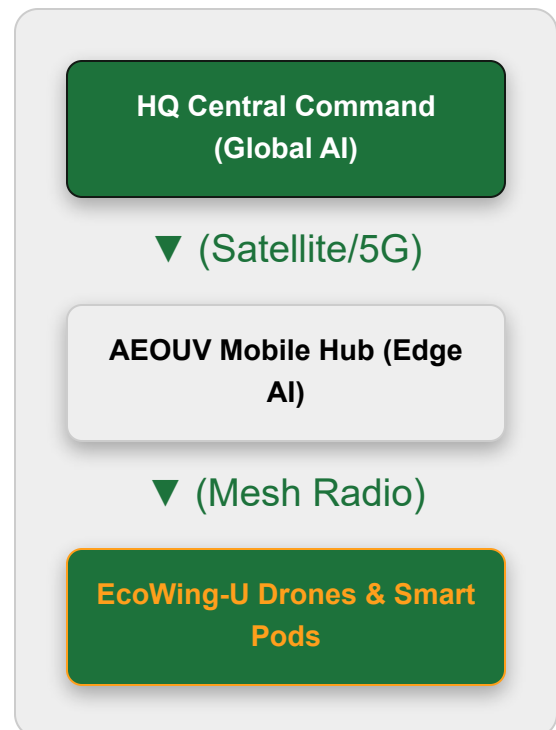


Swarm Symbiosis

Ground provides energy & computing.
Air provides reach & speed.

System Architecture (3-Tier Model)

- ✓ **Tier 1: Central HQ (Strategic AI):** High-level pathing, satellite data analytics, and regional drone-fleet coordination.
- ✓ **Tier 2: AEOUV Ground Hub (Edge AI):** Real-time self-driving path calculations, edge target-recognition, and local swarm management.
- ✓ **Tier 3: Aerial & Static Swarm (Task Execution):** VTOL drones executing tasks and tree-mounted Smart Pods forming a local sensor mesh.



AEOUV: Mobile Base Station & Lab

- ✓ **Environmental Diagnostics:** Integrated meteorological station monitoring real-time humidity, temperature, wind, CO₂, and dust metrics.
- ✓ **3D Perception:** Roof-mounted rotating LiDAR, stereo cameras, and mmWave radars for off-road autonomous trail navigation.
- ✓ **5-DOF Robotic Manipulator:** Collects soil cores and water samples directly from trails with automatic tool-changing end-effectors.
- ✓ **Power & Logistics:** High-capacity lithium battery pack that doubles as a mobile charging dock for three drones.



Edge Server + Carrier

Heavy processing of video streams and sensor payloads takes place directly on the vehicle.

EcoWing-U: Universal Conservation Drone

- ✓ **Hybrid VTOL Design:** Vertical rotors for ascending through tight forest canopies, switching to rear pusher propeller and folding wings for efficient 25 km forward cruise flight.
- ✓ **Single Manufacturing Line:** All drones share the identical airframe, flight computers, navigation sensors, and batteries, reducing spare parts overhead.
- ✓ **Full Mission Redundancy:** Drones have no fixed role. If Drone 1 is damaged, Drone 2 immediately clips onto the same Mission Pod to resume the task.



VTOL Carbon Fiber Airframe

Twin lift rotors, rear cruise propeller, and automatic hatch-docking interface.

Adaptive Mission Bay (AMB)

- ✓ **Automated Payload Swapping:** Fully enclosed, weatherproof internal compartment storing a magazine of six Mission Pods.
- ✓ **Robotic Linear Shuttle:** Replaces payloads in under 30 seconds without human hands.
- ✓ **Universal Interface:** Uses high-strength electro-permanent alignment magnets, pairing power and data contacts instantly.
- ✓ **Sealed Roof Hatches:** Roof hatches open only during launch/landing to keep internal electronics free from mud and rain.



Sub-30s Swap Time

Internal rail shuttle transfers pods automatically from vehicle magazine to drone.

Modular Mission Pods

1. Vaccination Pod

CO₂-driven micro-dart launcher and targeting optics for remote vaccine delivery to wildlife.

2. Seed Planting Pod

Drops biodegradable capsule seeds wrapped in nutrient shell and water-absorbent casing.

3. Sampling Pod

Lowers specialized siphon tubes into remote rivers and swamps to collect and preserve samples.

4. Emergency Pod

Drops emergency medical supplies, satellite signaling beacons, and walkie-talkies to ground rangers.

Smart Pods & Forest Sensor Mesh

- ✓ **Continuous Forest Auditing:** Low-power nodes deployed by drones onto tree branches or rivers.
- ✓ **Diverse Sensors:** Acoustic listening pods (detect gunshots/chainsaws), water quality probes, and motion-activated visual nodes.
- ✓ **Self-Healing Mesh Network:** Since canopy trees block standard signals, pods automatically route data packets hop-by-hop back to the AEUV.



Self-Configuring Network

Pods act as relay repeaters, removing communication blindspots in deep valleys.

System ROI & Positioning

- ✓ **Operational Cost Reduction:** Shifting patrol operations from multiple high-maintenance diesel jeeps to a single electric ground hub and drones.
- ✓ **Zero Soil & Habit Damage:** Eliminates off-road vehicle driving, stopping soil erosion and animal migration routing disruption.
- ✓ **Auditable Conservation:** High-resolution seed placement maps and water logs provide immediate audits for carbon credit and NGO funding.



Competitive Advantage

ACS is positioned as a comprehensive conservation platform, not just a hardware ATV.